Network Layer Security_IPSec
Module 3

Prepared by

Jesna J S

Assistant Professor, CSE TKMCE

Topics Includes:

- Internet Protocol Security (IPSec) – Overview,
- IP security architecture,
- Authentication Header (AH),
- Encapsulating Security Payload (ESP),
- Combining Security Associations,
- Key management : Internet Key Exchange (IKE) - Phases.

Internet Protocol Security[IPSec]

- IP Security (IPSec) is a collection of protocols designed by the Internet Engineering Task Force (IETF).
- IPSec helps **create authenticated and confidential packets for the IP layer**

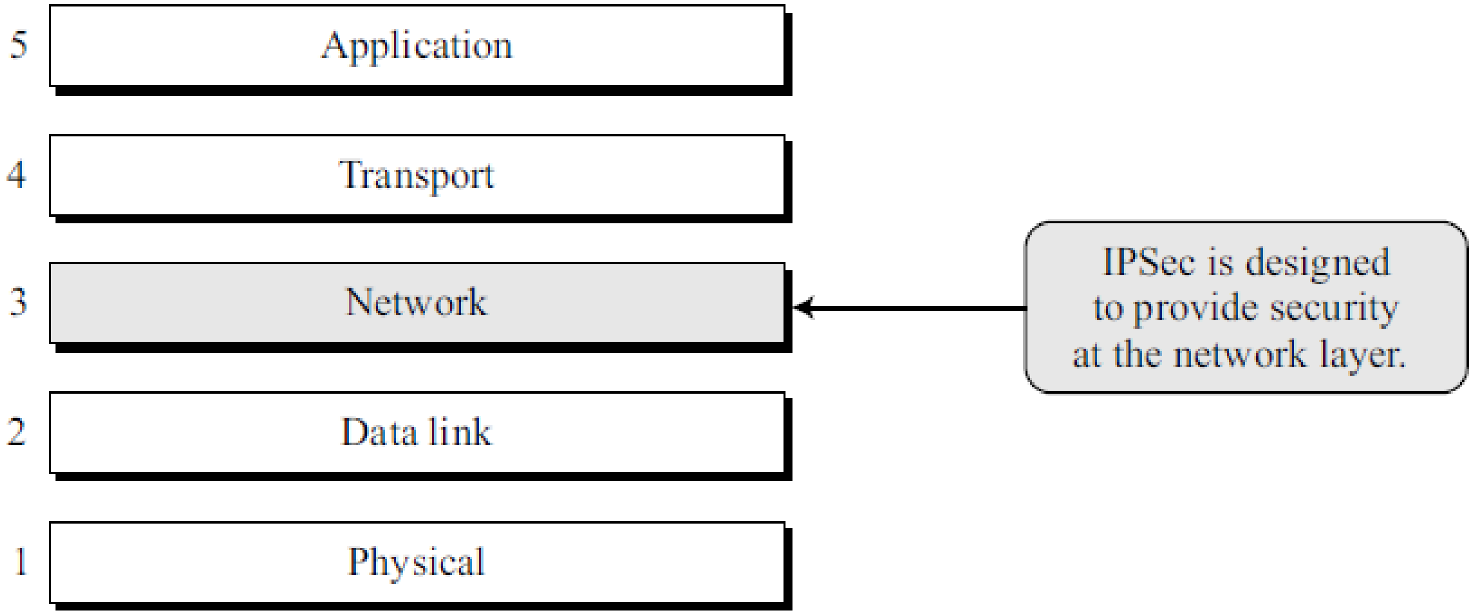
Objectives:

- Improve security for **client/server programs such as electronic mail**.
- Improve security for **client/server programs such as HTTP**.
- Add security to client/server applications **that do not use transport-layer security**.
- Provide security for **node-to-node communication, such as routing protocols**.

Operating modes of IPSec

- IPSec operates in one of two different modes:
    1. **Transport mode**
    2. **Tunnel mode.**

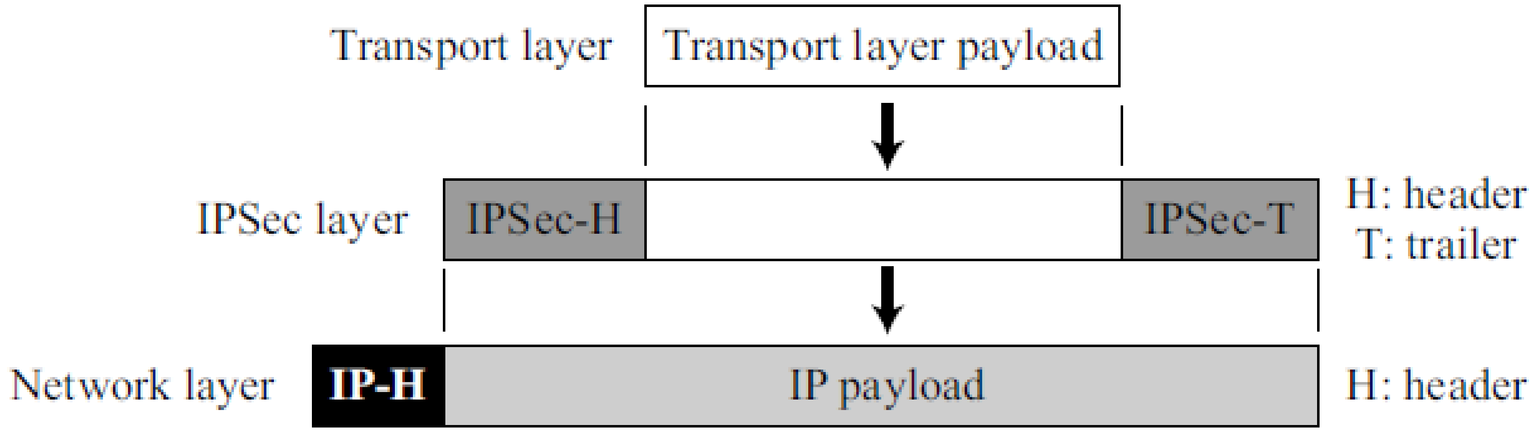
18.1 *TCP/IP protocol suite and IPSec*



1. Transport Mode:

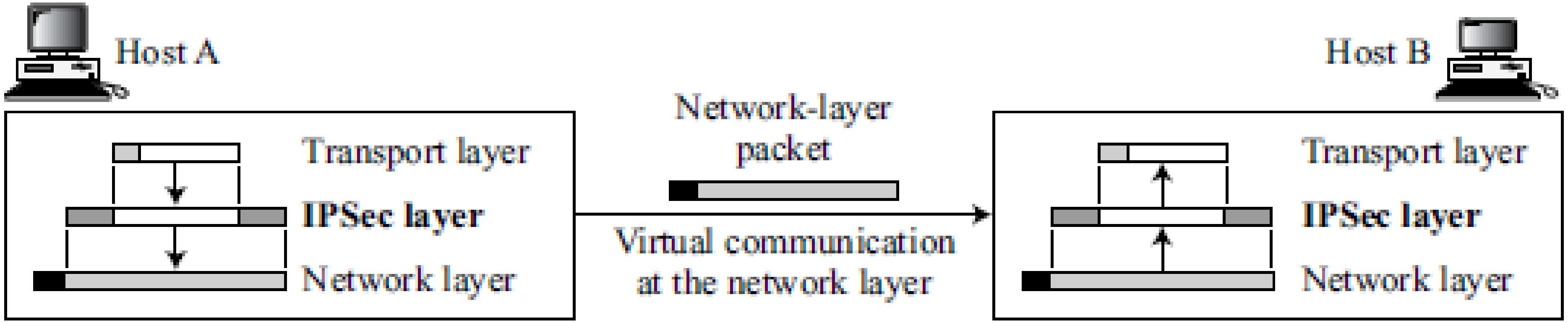
- Transport mode **protects only the transport-layer payload**, not the IP header.
- The **original IP header remains unchanged and unprotected**.
- IPSec header and trailer are added to the data received from the transport layer.
- **Encryption and/or authentication is applied only to the IP payload.**
- The IP header is added after IPSec processing

18.2 *IPSec in transport mode*



- Used mainly for **host-to-host (end-to-end) communication**.
- **Sender encrypts/authenticates the payload**;
- **receiver decrypts/verifies and passes it to the transport layer**.
- Suitable when both sender and receiver are hosts.

Figure 18.3 *Transport mode in action*



2. Tunnel Mode:

- Tunnel mode **protects the entire IP packet**, including the original IP header.
- **The complete IP packet is encrypted/authenticated first.**
- A **new IP header is added** outside the protected packet.
- The new IP header contains **different addressing information**.
- Commonly used between **routers, host-to-router, or router-to-host**.
- Provides security as if the packet travels through a **secure tunnel**.
- Suitable when **either the sender or receiver is not a host**.

IPSec in tunnel mode

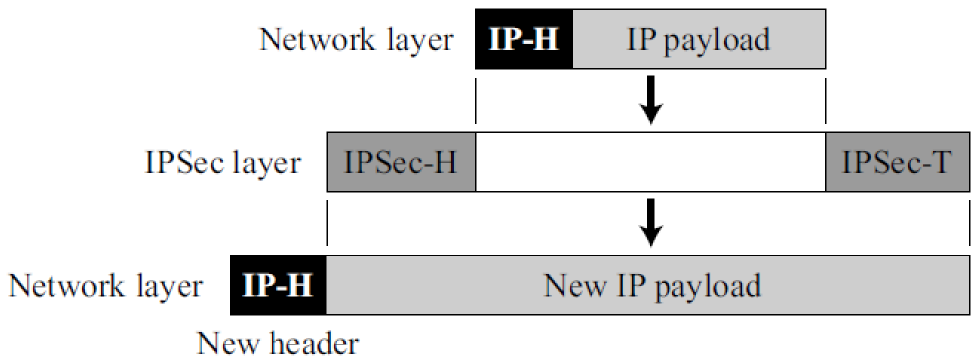
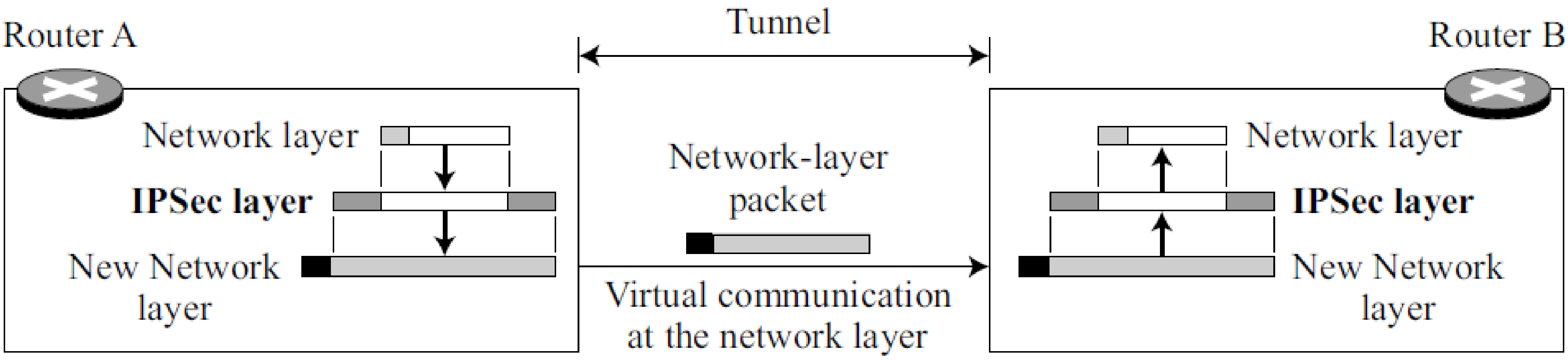


Figure 18.5 *Tunnel mode in action*



packet is protected from intrusion between the sender and the receiver, as if the whole packet goes through an imaginary tunnel.

IPSec in tunnel mode protects the original IP header.

IPSec Security Protocols

- IPSec defines two protocols to provide **authentication and/or encryption for packets at the IP level.**
    1. **Authentication Header (AH) Protocol**
    2. **Encapsulating Security Payload (ESP) Protocol**

Authentication Header (AH) Protocol

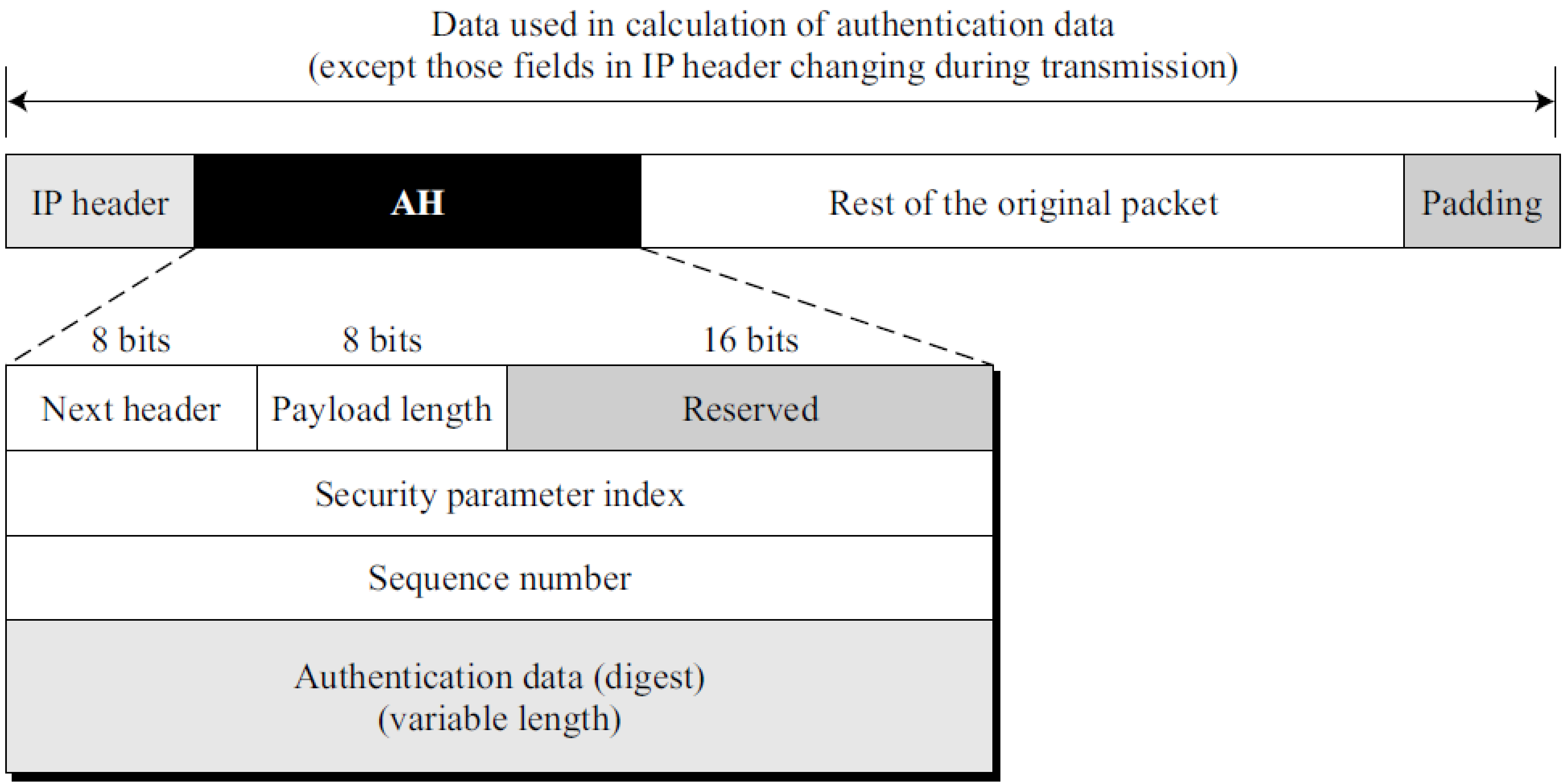
- AH is used to **authenticate the source host**.
- It ensures **data integrity of the IP packet**.
- AH **does not provide encryption** (no confidentiality).
- **Uses a hash function with a shared symmetric key to create a message digest.**
- A message digest is computed over:
  - ✓The IP packet payload
  - ✓Immutable parts of the IP header (fields that do not change in transit)
- This **digest is placed in the Authentication Data field of AH**.
- At the receiver:
  - ➢ The hash is recomputed using the same key.
  - ➢If both digests match, data integrity and source authentication are confirmed

- AH is placed in the IP packet based on the **mode used**:
  - ✓**Transport mode**: **AH protects the transport-layer payload**.
  - ✓**Tunnel mode**: **AH protects the entire IP packet**.
- When AH is used, the Protocol field in the IP header is set to 51.
- The original protocol value (TCP, UDP, etc.) is stored in the Next Header field of AH.

Steps in Adding Authentication Header

1. An authentication header is added to the payload with the authentication data field set to 0.
2. Padding may be added to make the total length even for a particular hashing algorithm.
3. Hashing is based on the total packet. However, only those fields of the IP header that do not change during transmission are included in the calculation of the message digest (authentication data).
4. The authentication data are inserted in the authentication header.
5. The IP header is added after changing the value of the protocol field to 51.

Figure 18.7 *Authentication Header (AH) protocol*



Fields of Authentication Header

1. Next Header (8 bits)

- Identifies the **type of payload** (TCP, UDP, ICMP, etc.).
- Contains the **original protocol field value** of the IP header.

2. Payload Length (8 bits)

- Indicates the **length of the AH in 4-byte units**.
- Does **not include the first 8 bytes** of AH.

3. Security Parameter Index (SPI) – 32 bits

- Identifies the **Security Association (SA)**.
- Acts like a **virtual circuit identifier**.
- Same SPI is used for all packets in one SA.

4. Sequence Number – 32 bits

- Provides **packet ordering**.
- Prevents **replay attacks**.
- Sequence number:
  - Is never repeated
  - Does not wrap around after 2^{32}
  - Requires a new SA when exhausted

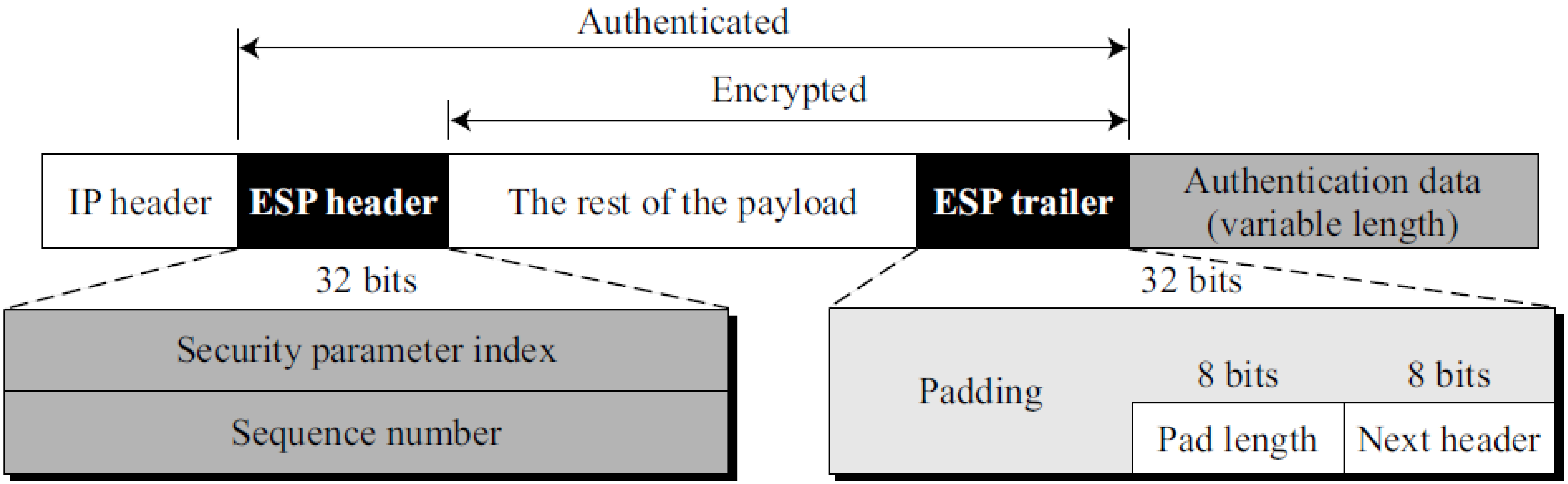
5. Authentication Data

- Contains the **hash-based message digest**.
- Ensures **data integrity and source authentication**.
- Calculated over the IP packet excluding mutable fields (e.g., TTL).

Encapsulating Security Payload (ESP)

- ESP provides **confidentiality (privacy), data integrity, and source authentication**.
- It overcomes the limitation of AH by encrypting the data.
- ESP **adds both a header and a trailer to the IP packet.**
- ESP inserts:
  - ✓An ESP header before the payload
  - ✓An ESP trailer after the payload
- Authentication data is added at the end of the packet, simplifying its calculation.
- When ESP is used, the Protocol field in the IP header is set to 50.
- The original protocol value (TCP, UDP, etc.) is stored in the Next Header field in the ESP trailer.

Figure 18.8 *ESP*



ESP Processing Steps:

1. ESP trailer is added to the payload.
2. The payload and ESP trailer are encrypted.
3. The ESP header is added.
4. Authentication data is calculated over the ESP header, encrypted payload, and trailer.
5. Authentication data is appended at the end of the ESP trailer.
6. IP header is added with Protocol value = 50.

Fields of ESP Header and Trailer

- **Security parameter index.** The 32-bit security parameter index field is similar to that defined for the AH protocol.
- **Sequence number.** The 32-bit sequence number field is similar to that defined for the AH protocol.
- **Padding.** This variable-length field (0 to 255 bytes) of 0s serves as padding.
- **Pad length.** The 8-bit pad-length field defines the number of padding bytes. The value is between 0 and 255; the maximum value is rare.
- **Next header.** The 8-bit next-header field is similar to that defined in the AH protocol. It serves the same purpose as the protocol field in the IP header before encapsulation.
- **Authentication data.** Finally, the authentication data field is the result of applying an authentication scheme to parts of the datagram. Note the difference between the authentication data in AH and ESP. In AH, part of the IP header is included in the calculation of the authentication data; in ESP, it is not.

Services Provided by IPSec

Table 18.1 *IPSec services*

Services	*AH*	*ESP*
Access control	yes	yes
Message authentication (message integrity)	yes	yes
Entity authentication (data source authentication)	yes	yes
Confidentiality	**no**	yes
Replay attack protection	yes	yes

Security Association

- A Security Association (SA) is a **logical agreement between two parties**.
- **It defines how security is applied to IP packets.**
- An SA **creates a secure communication channel**.
- Security Associations **are unidirectional**.
- For two-way communication, two SAs are required:
  - ✓Outbound SA (sender → receiver)
  - ✓Inbound SA (receiver → sender)

Information Stored in a Security Association : An SA may include:

- ✓**Encryption algorithm and key**
- ✓**Integrity/authentication algorithm and key**
- ✓**IPSec protocol used (AH or ESP)**
- ✓**Other security parameters**

Table 18.2 *Typical SA Parameters*

Parameters	*Descriptions*
Sequence Number Counter	This is a 32-bit value that is used to generate sequence numbers for the AH or ESP header.
Sequence Number Overflow	This is a flag that defines a station's options in the event of a sequence number overflow.
Anti-Replay Window	This detects an inbound replayed AH or ESP packet.
AH Information	This section contains information for the AH protocol: 1. Authentication algorithm 2. Keys 3. Key lifetime 4. Other related parameters
ESP Information	This section contains information for the ESP protocol: 1. Encryption algorithm 2. Authentication algorithm 3. Keys 4. Key lifetime 5. Initiator vectors 6. Other related parameters
SA Lifetime	This defines the lifetime for the SA.
IPSec Mode	This defines the mode, transport or tunnel.
Path MTU	This defines the path MTU (fragmentation).

- If only confidentiality is required:
    - ✓**Both parties share a secret key.**
    - ✓**They agree on an encryption/decryption algorithm.**
    - ✓**Sender encrypts data using the key and algorithm.**
    - ✓**Receiver decrypts data using the same key and algorithm.**
- For authentication and integrity, additional parameters are included.
- Separate SAs may exist for AH and ESP, each with different algorithms and keys.

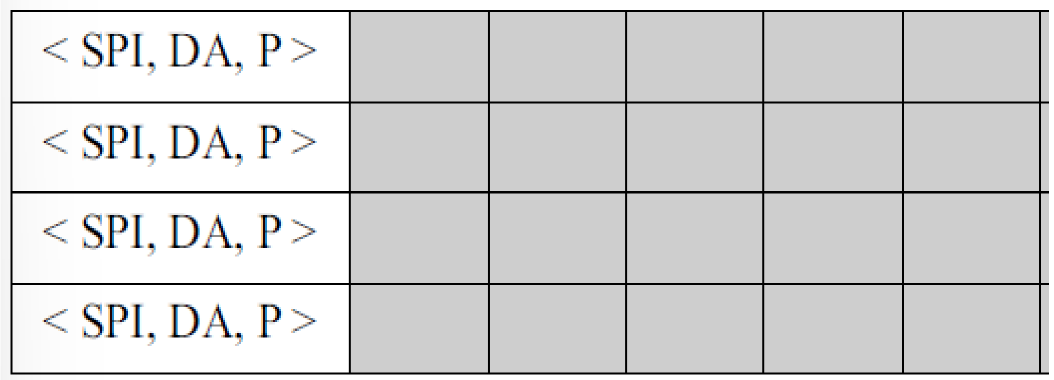
Security Association Database (SAD)

- Security Association Database (SAD) **stores all active SAs**.
- **Each entry in the SAD represents one Security Association.**
- It can be visualized as a table of SAs.
- **Types of SAD**
  1. **Outbound SAD**: Used when sending IPSec packets.
  2. **Inbound SAD**: Used when receiving IPSec packets.
- **Use of SAD**
  - ✓While **sending**, the host searches the **outbound SAD** to apply security.
  - ✓While **receiving**, the host searches the **inbound SAD** to verify security.
  - ✓Correct SA selection is critical for proper packet processing.

Indexing an Inbound SAD Entry

- Each inbound SA is uniquely identified using a **triple index**:

1. **Security Parameter Index (SPI)**
   - ✓A **32-bit value** that identifies the SA.
   - ✓Assigned during **SA negotiation**.
   - ✓Same SPI is used for all packets of that SA.



Security Association Database

2. **Destination Address**
   - ✓Identifies the **receiving host**.
   - ✓Separate SAs are required for each **destination address**.
   - ✓Supports both **unicast and multicast** addresses.

3. **Protocol**
   - ✓Specifies the IPSec protocol used:
     - **AH**
     - **ESP**
   - ✓Separate SAs are maintained for AH and ESP.